

Exercise

Background: You are the chief health officer in the isolated city-state of SMDMopolis, which has its first ever documented case of **Virus X**. Virus X is an emerging infectious disease, and you have been tasked with projecting outbreak trends and identifying a policy response for the 10,000 residents of SMDMopolis.

Part 1: You review the limited literature from a handful of past outbreaks in other locations and find:

- People tend to feel better 14 days after developing symptoms.
 - In previous outbreaks, contact tracing studies have found that the average infected person infected between 3 and 10 other people.
1. Based on the information provided in the prompt, can you estimate best-case and worst-case effective contact rates (β) for Virus X?
 2. Combine your answer to part A with an SIR model to describe, under best- and worse-case scenarios:
 - a. The cumulative number of infections at 300 days (absent any intervention)
 - b. The likely size and timing of the epidemic peak (i.e., on which day is prevalence the highest, and how many new infections will that be?)

Bonus questions (if time permits)

3. Plot:
 - a. The epidemic trace curve, showing the number of people who are susceptible, infected, and recovered over time.
 - b. Cumulative infections over time.

Part 2: In previous outbreaks, which tended to last 1-2 months, there were no documented instances of someone acquiring Virus X twice. However, a study has just been published which found that antibody levels wane substantially 3 months after recovery.

1. Update the model to incorporate waning immunity.
2. Incorporate this new data into your updated model.
 - a. Remember to keep your time scales (days, weeks, months) consistent.
3. How do your updates change your answers to exercise part 1? Consider again the best-case and worst-case scenarios.
 - a. The cumulative number of infections at 300 days (absent any intervention)
 - b. The likely size and timing of the epidemic peak
 - c. Describe your intuition for the directions and/or magnitude of changes.

Bonus questions (if time permits)

4. Does this update affect estimates of the basic reproductive number? How about the effective reproductive number?
 - a. Note: you do not have to recalculate any reproductive numbers – just answer yes/no and think about your reasoning.
5. You have also learned from SMDMopolis' chief demographer that last year, there were around 120 new births and 120 deaths documented in the population. Given the 300 day projections period you are considering, how important is it to incorporate demographics in your model? Would your answer change if you were modeling over a longer timeframe (say, 10 years)?
 - a. Note: we are not asking you to update your model here – just consider the question and formulate a response.

Part 3: Two vaccines are available that provide partial immunity against Virus X. Vaccine A is 65% effective and has negligible side effects, so you the initial¹ vaccination rate will be 150 people per day. Vaccine B is 90% effective but has more side effects, so estimated coverage is lower: only 50 people per day initially. Both vaccines provide on average 1 year of protection.

For simplicity you can assume that protection begins immediately, that vaccination does not reduce the infectiousness of breakthrough infections, and that only susceptibles receive the vaccine.

1. Compare cumulative numbers of infections between the two vaccine options. Consider both best- and worst-case scenario of viral transmission. You can start from the vaccination functions in `SIRS_closed_vax.R` if you find this helpful rather than building your own function from scratch.

You would like to consider cost-effectiveness as an input to your decision about which vaccine option to choose. You are given the following information:

	Vaccine A	Vaccine B
Costs		
Cost per dose	\$100	\$30
Fixed cost of starting a vaccine program	\$10,000	\$10,000
Cost per day spent with Virus X (productivity)	\$50	
QALY Weights		
S, R, V states	1.0 per year spent in the state	
I state	0.75 per year spent in the state	

¹ Note that in these ODE-based compartmental models, transition rates are defined per person in the compartment, so as the number of susceptibles declines so will the number of people moving from S to V for a given fixed vaccination rate. In practice, you would need to set up a time-varying parameter to allow a constant number for people to be vaccinated each day. For this exercise, you can ignore this nuance.

Vaccine side effects	None	0.95 per year, side effects last 3 days
----------------------	------	--

2. Report the total costs and QALYs (quality-adjusted life years) of each vaccine option and of the no vaccine base case, accrued over 300 days after the outbreak.
3. Estimate incremental cost-effectiveness ratios (ICERs) as incremental costs divided by incremental QALYs, and make a decision at willingness-to-pay thresholds of \$5,000/QALY and \$10,000/QALY. For the time horizon considered here, you may ignore any discounting of future costs and QALYs.

Bonus question (if time permits)

4. How does your answer to #3 change if vaccine A also reduces infectiousness by 50%?